

NAME:- VIBHU DIXIT

Roll No:- 2215001940 (77)

Date: ___/___/___

SECTION:- B

MON	TUE	WED	THR	FRI	SAT	SUN
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Subject

5/2/2025

ASSIGNMENT 1: DIGITAL IMAGE PROCESSING

①

Given:- Image dimensions = 128×128
 Bands = 3 (RGB)
 Bits per pixel = 8

To find:- Total memory = ?
 Total memory (Grayscale) = ?

Solⁿ

$$\begin{aligned}
 \text{Total Memory} &= \text{Width} \times \text{Height} \times \text{Bands} \times \text{Bits per pixel} \div 8 \\
 &= 128 \times 128 \times 3 \times 8 \div 8 \\
 &= 128 \times 128 \times 3 \\
 &= 49,152 \text{ bytes} \\
 &= 48 \text{ KB}
 \end{aligned}$$

If the image is converted to grayscale, there is only one band instead of three:

$$\begin{aligned}
 \text{Total Memory (Grayscale)} &= 128 \times 128 \times 1 \times 8 \div 8 \\
 &= 16,384 \text{ bytes} \\
 &= 16 \text{ KB}
 \end{aligned}$$

∴ RGB Image Memory = 48 KB

∴ Grayscale Image Memory = 16 KB

∴ Memory Reducⁿ = 67% decrease (1/3rd of original size)

②

(i) $V = \{2, 3\}$

Pixels with values 2 and 3 are considered connected!!

• 4-Connectivity (D4)

A pixel is connected to its immediate horizontal & vertical neighbors

Possible moves: (UP, Down, Left, Right)

```

  *
 * P *
  *

```

Valid paths for $V = \{2, 3\}$

2(P)	→ 3	→ 2	↓	6	1
6	2	3	↓	6	2
5	3	2	↓	3	5
2	4	3	↓	5	2
4	5	2	→ 3	→ 6(Q)	

Here gray level of pixel 'q' = 6
 which is not belongs to 'V'.

↓
 So there is no path through
 p & q are connected!!

• 8-Connectivity (D8)

A pixel connected to all eight neighbors (horizontal, vertical & diagonal)

```

  * * *
 * P *
  * * *

```

Here also 'q' = 6 not belongs to 'V'

• m-Connectivity (Dm)

Subject _____

(ii) $V = \{2, 6\}$ • 4- Connectivity (D_4)

2(p)	3	2	6	1
6	2	3	6	2
5	3	2	3	5
2	4	3	5	2
4	5	2	3	6(q)

No path x

• 8- Connectivity (D_8)

2(p)	3	2	6	1
6	2	3	6	2
5	3	2	3	5
2	4	3	5	2
4	5	2	3	6(q)

No path

$$D_4(p, q) = |x-s| + |y-t| = |0-5| + |0-5| = 5+5 = 10 \text{ units}$$

$$D_8(p, q) = \max(|x-s|, |y-t|) = \max(|0-5|, |0-5|) = 5 \text{ units}$$

D_m distance b/w two pixels depends on the values of the pixels along with the path & also on the values of their neighbours

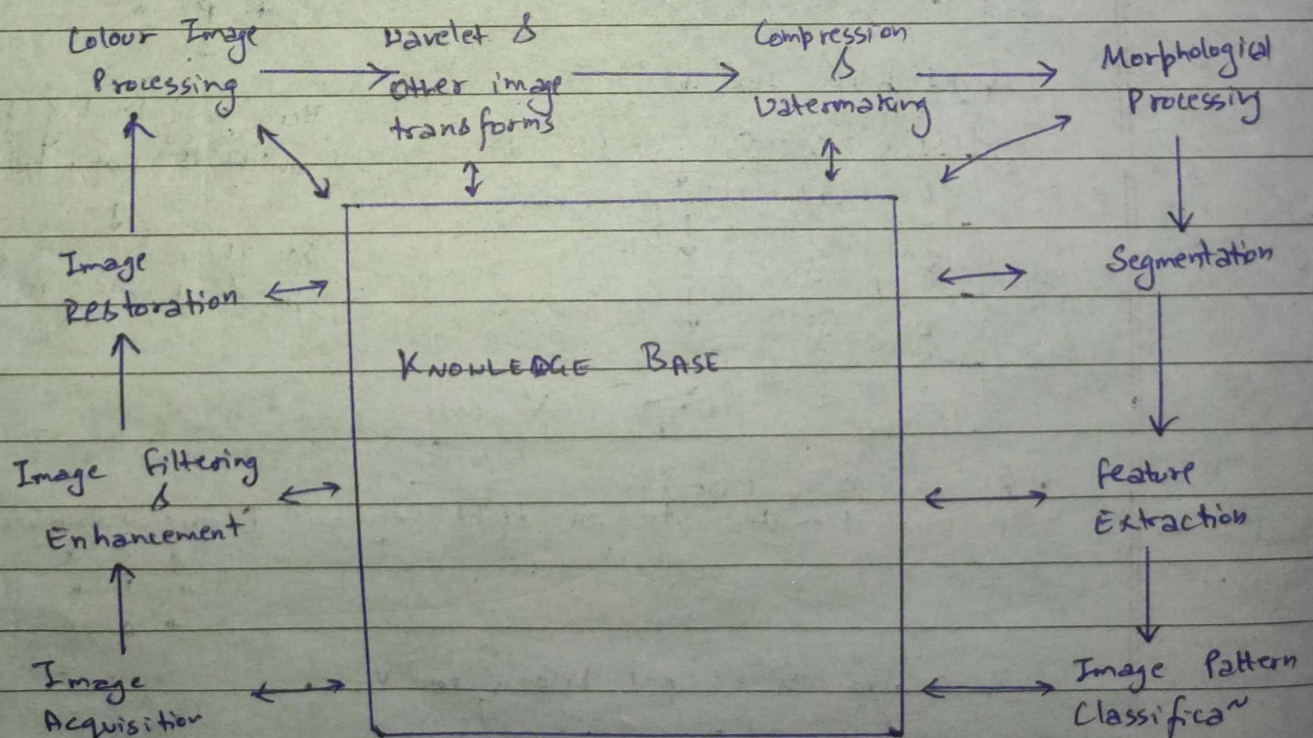
(i) $V = \{2, 3\}$

2(p)	3	2	6	1
6	2	3	6	2
5	3	2	3	5
2	4	3	5	2
4	5	2	3	6(q)

(ii) $V = \{2, 6\}$

2(p)	3	2	6	1
6	2	3	6	2
5	3	2	3	5
2	4	3	5	2
4	5	2	3	6(q)

3



4

Given: Image size = 1024×1024 pixel
 Gray levels = 256

Each packet :- Start bit = 1 bit
 Data byte = 8 bits
 Stop bit = 1 bit
 Total bits per pixel = $1+8+1$
 $\rightarrow 10$ bits

Modem Speeds :- 750k baud
 56k baud

(a) Time over a 750k baud modem

Total pixels in image = 1024×1024

= 1,048,576 pixels

Total bits to transmit = $1,048,576 \times 10$
 = 10,485,760 bits

Transmission time = $\frac{\text{Total bits to transmit}}{\text{Modem speed}} = \frac{10,485,760 \text{ bits}}{750,000 \text{ bits/sec}} = 13.98 \text{ sec} \approx 14 \text{ sec}$

(b) Time over a 56k baud modem

Total bits to transmit = 10,485,760 bits

Transmission time = $\frac{10,485,760}{56,000} = 187.25 \text{ sec.}$

Convert sec to min = $\frac{187.25}{60} \approx 3.12 \text{ minutes}$

5

(a) $V = \{0, 1, 2\}$

• 4-path does not exist

• 8-path does not exist

• M-path does not exist

3	4	1	2	0
0	1	0	4	2 (u)
2	2	3	1	4
(e) 3	0	4	2	1

(b) $V = \{2, 3, 4\}$

• 4-path does not exist

• 8-path : 4 (3-2-3-4-2)

• m-path : 5 (3-2-2-3-4-2)

3	4	1	2	0
0	1	0	4-2 (u)	
2	2	3	1	4
1	3	0	4	2
3				1

⑥

Given: Image size = 100×100
 $= 10,000$ pixels

Each pixel can have 2 possible values (0 or 1)

Sol: Total no. of distinct binary images = $2^{10,000}$

⑦

Given: Image resolution = 1024×1024

Gray levels = 64

layers = 100

Transfer rate = 128 k bits/sec

Packet size = 100 bits

start & stop flag = 14 bits each

(a) Storage Req.

Total pixels per image = 1024×1024
 $= 1,048,576$ pixels

Bits per image = $1,048,576 \times 6$
 $= 6,291,456$ bits

Total bits for 100 layers = $6,291,456 \times 100$

$= 629,145,600$ bits

Convert bits to GB = $\frac{629,145,600}{8} = 78,643,200$ bytes = 0.0786 GB

(b) Transfer Time Calculatⁿ

Total bits to transmit = 629,145,600 bits

Bits per packet = $100 + 28 = 128$ bits per packet

No. of packets req. = $\frac{629,145,600}{100}$
 $= 6,291,456$ packets

Total bits including start/stop flag = $6,291,456 \times 128$
 $= 804,924,408$ bits

Transmission time in sec = $\frac{804,924,408}{128,000} = 6,289$ sec.

Convert to hours = $\frac{6,289}{3600} \approx 1.75$ hours

⑧

Types Of Image Resolun

1. Spatial Resdution

↳ Refers to the no. of pixels in an image

↳ A spatial resoluⁿ means more details & sharpness

↳ Ex. A 1080p image has higher spatial resoluⁿ than a 720p

2. Gray Level Resoluⁿ (Bit Depth)

↳ Represents the no. of intensity levels per pixel

↳ A higher bit depth provides smoother gradations in brightness

↳ Ex.

8-bit image → 256 gray levels

12-bit image → 4096 gray levels

3. Spectral Resolⁿ

- No. of color bands captured
- More spectral bands allow for better color differentiaⁿ

4. Temporal Resolⁿ

- No. of images captured per sec.
- Higher temporal resolⁿ is crucial for video recording & real-time processing

Choice:- If the goal is general usage (photos, videos & display), a mobile with higher resolⁿ is preferable.
If medical imaging or professional photography is needed, higher gray level resolⁿ is beneficial !!

(9)

	0	1	2	3
(P) 0	1	0	0	0
1	1	1	0	0
2	1	1	1	1
3	1	0	1	1 (Q)

Shortest length = 6

Path:- 1 → 1 → 1 → 1 → 1 → 1

$$P(x, y) = (0, 0)$$

$$P(s, t) = (3, 3)$$

(10)

$$\begin{aligned} \text{Euclidean Distance } (D_e) &= \sqrt{(s-x)^2 + (t-y)^2} \\ &= \sqrt{(3-0)^2 + (3-0)^2} \\ &= \sqrt{18} \end{aligned}$$

$$= 4.24 \text{ units}$$

$$\begin{aligned} \text{City Block Distance } (D_4) &= |s-x| + |t-y| \\ &= |3-0| + |3-0| \\ &= 6 \text{ units} \end{aligned}$$

$$\begin{aligned} \text{Chessboard Distance } (D_8) &= \max(|s-x|, |t-y|) \\ &= 3 \text{ units} \end{aligned}$$